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Illuminated electrosurgical devices, systems and methods

Abstract

Embodiments relate to illuminated electrosurgical devices and related systems and methods. An electrosurgical device can include an electrosurgical blade having a proximal end and a distal end and comprising a dielectric coating, at least one of the electrosurgical blade or the dielectric coating including at least one light-guiding element, an optic having a proximal end and a distal end, the distal end of the optic optically coupled to the proximal end of the electrosurgical blade, and a light source arranged at the proximal end of the optic and configured to provide light that is guided through the optic and the at least one light-guiding element to illuminate an area around the distal end of the electrosurgical blade.

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Background/Summary

RELATED APPLICATION (1) This application claims the benefit of U.S. Provisional Application No. 62/660,396 filed on Apr. 20, 2018, the entirety of which is incorporated herein by reference.

TECHNICAL FIELD

(1) This disclosure relates generally to electrosurgical devices and more particularly to electrosurgical devices with integrated illumination systems.

BACKGROUND

(2) Electrosurgical devices generally include a handpiece (handle) ergonomically adapted for ease of manipulation by a surgeon during surgery, and for positioning an energy tip of the device to deliver electrical energy to a target tissue for tissue cutting or coagulation. An electrode and electrical supply cable are generally disposed within the handpiece, traversing from the handpiece's proximal end through the handpiece body, and terminating in an energy discharge tip at the distal end of the device. The electrical supply cable typically is connected to an energy source, such as a radiofrequency (RF) energy generator.

(3) The handpiece or other portion of the device may include an illumination element for illuminating the surgical field. Light may be conducted towards the energy discharge tip and directed onto the surgical field via an optical waveguide coupled to the handpiece or disposed within the handpiece. The electrode may be disposed within the optical waveguide, or disposed alongside the waveguide. The electrode and waveguide may be disposed within a suitable supporting structure (for example, a cylindrical metal tube), that may be slidably extendable or retractable to permit the electrosurgical device to elongate or shorten as needed to treat the surgical site.

SUMMARY

(4) Embodiments relate to illuminated electrosurgical devices and related systems and methods.

(5) In one embodiment, an electrosurgical device comprises an electrosurgical blade having a proximal end and a distal end and comprising a dielectric coating, at least one of the electrosurgical blade or the dielectric coating including at least one light-guiding element, an optic having a proximal end and a distal end, the distal end of the optic optically coupled to the proximal end of the electrosurgical blade, and a light source arranged at the proximal end of the optic and configured to provide light that is guided through the optic and the at least one light-guiding element to illuminate an area around the distal end of the electrosurgical blade.

(6) In another embodiment, a method comprises coating an electrosurgical blade having a proximal end and a distal end with a dielectric layer, forming at least one light-guiding element on at least one of the electrosurgical blade or the dielectric layer, optically coupling a distal end of an optic to the proximal end of the electrosurgical blade, and arranging a light source at a proximal end of the optic to provide light that is guided through the optic and the at least one light-guiding element of the dielectric layer to illuminate an area around the distal end of the electrosurgical blade.

(7) The above summary is not intended to describe each illustrated embodiment or every implementation of the subject matter hereof. The figures and the detailed description that follow more particularly exemplify various embodiments.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) Subject matter hereof may be more completely understood in consideration of the following detailed description of various embodiments in connection with the accompanying figures, in which:

(2) FIG. 1 depicts an electrosurgical device according to an embodiment.

(3) FIG. 2 depicts an internal view of the electrosurgical device of FIG. 1.

(4) FIG. 3 depicts a blade assembly of the electrosurgical device of FIG. 1.

(5) FIG. 4 depicts an optic and LED of the blade assembly of FIG. 3.

(6) FIG. 5 depicts a detail view of an optical coupling of the blade assembly of FIG. 3.

(7) FIG. 6 depicts a detail view of a distal tip of a blade of the blade assembly of FIG. 3.

(8) FIG. 7 depicts a blade of an electrosurgical device according to an embodiment.

(9) FIG. 8 depicts blade coating patterns.

(10) FIG. 9 depicts a cross-sectional view of a portion of a blade of an electrosurgical device according to an embodiment.

(11) FIG. 10 depicts another cross-sectional view of a portion of a blade of an electrosurgical device according to an embodiment.

(12) FIG. 11 depicts a detail view of a distal tip of a blade of an electrosurgical device according to an embodiment.

(13) FIG. 12 depicts another detail view of a distal tip of a blade of an electrosurgical device according to an embodiment.

(14) FIG. 13 depicts an exploded view an electrosurgical device according to an embodiment.

(15) FIG. 14 depicts an exploded view of a blade assembly of the electrosurgical device of FIG. 13.

(16) FIG. 15 depicts a first side of a Fresnel lens according to an embodiment.

(17) FIG. 16 depicts a second side of the Fresnel lens of FIG. 15.

(18) FIG. 17 depicts a side view of the Fresnel lens of FIG. 15.

(19) FIG. 18 depicts another view of the Fresnel lens of FIG. 15.

(20) FIG. 19 depicts a cross-sectional view of the Fresnel lens of FIG. 15, taken along line A-A in FIG. 18.

(21) FIG. 20 depicts a cable assembly of the electrosurgical device of FIG. 13.

(22) FIG. 21 depicts an exploded view of the cable assembly of the electrosurgical device of FIG. 13.

(23) FIG. 22 depicts an internal view of an electrosurgical device according to an embodiment.

(24) FIG. 23 depicts an exploded view of the electrosurgical device of FIG. 22.

(25) FIG. 24 depicts another view of the electrosurgical device of FIG. 22.

(26) FIG. 25 depicts an exploded view an electrosurgical device according to an embodiment.

(27) FIG. 26 depicts another view of the electrosurgical device of FIG. 22.

(28) FIG. 27 depicts a lens for an electrosurgical device according to an embodiment.

(29) FIG. 28 depicts another lens for an electrosurgical device according to an embodiment.

(30) While various embodiments are amenable to various modifications and alternative forms, specifics thereof have been shown by way of example in the drawings and will be described in detail. It should be understood, however, that the intention is not to limit the claimed inventions to the particular embodiments described. On the contrary, the intention is to cover all modifications, equivalents, and alternatives falling within the spirit and scope of the subject matter as defined by the claims.

DETAILED DESCRIPTION OF THE DRAWINGS

(31) Unless the context indicates otherwise, the following terms shall have the following meaning and shall be applicable both to the singular and plural:

(32) The term “electrosurgical device” means an electrical device designed for handheld use by a surgeon to dispense RF or other energy through the tip of an electrode into target surgical tissue, in

order to cut or coagulate the tissue during a surgical procedure.

(33) The terms “insulator,” “insulation” and “insulating” mean electrically insulating, and refer to dielectric materials that permit little, if any, flow of electrical current through such material. Insulating materials may in some instances be thermal insulators but are not always so. Materials such as glass, metal oxides, porcelain, paper, plastics, polymers and rubbers are representative insulating materials.

(34) The terms “radiofrequency energy” or “RF energy” mean energy from the electromagnetic spectrum having a frequency between about 3 kilohertz (3 kHz) and about 300 gigahertz (300 GHz).

(35) The term “proximal” or “proximate,” in the context of an area or end of a device or element means the operator end of the device or element, while the term “distal” means the patient end of the device or element.

(36) Surgical devices should not unduly impede the surgeon's view of the operating field. This can be particularly troublesome in electrosurgical devices, especially those with extra features beyond energy delivery, such as added illumination, smoke evacuation, saline delivery, an extendable or rotatable shaft, a bendable tip, or other ancillary features.

(37) In the case of an electrosurgical device that also provides added illumination (viz. light directed at the surgical field), the light desirably is emitted near the distal end of the device, where any added bulk may also directly impede the surgeon's view. Device designers have consequently sought to minimize the distal profile of such devices, and to make the associated components as small, thin and few in number as possible. At the same time, device designers have sought to meet an expressed desire of surgeons for as much light as possible provided at the point of dissection.

(38) Referring to FIGS. 1 and 2, an embodiment of an electrosurgical device **100** is depicted. Generally, electrosurgical device **100** comprises a handpiece **102**, blade assembly **104** and switch assembly **106**. Handpiece **102** provides both a comfortable handle for a user to grip during use of device **100** and a housing for blade assembly **104** and switch assembly **106**. In some embodiments, handpiece **102** can be coupled to an external power source via an electrical cable (not shown in FIG. 1 or 2) or house a power source, such as one or more batteries (also not shown in FIG. 1 or 2).

(39) Other general features of electrosurgical device **100** are described in U.S. Pat. No. 7,736,361, US 2017/0172646 and US 2016/0120592, which are incorporated herein by reference to the extent each is consistent with the instant disclosure.

(40) Referring to FIGS. 3 and 4, blade assembly **104** comprises a light source **108**, an optic **110** and a blade **112**. Light source **108** can comprise a light emitting diode (LED) light source, incandescent light source, or other suitable type of light source in various embodiments. Light source **108** can comprise a single light element (e.g., a single LED) or a plurality of light elements (e.g., multiple LEDs). Operation of light source **108** can be controlled by switch assembly **106** such that light source **108** can be selectively turned on or off. In some embodiments, additional control of light source **108** or a characteristic thereof (e.g., direction, brightness, size) can be controlled by switch assembly **106** or another component of device **100**.

(41) Light source **108** is arranged proximate optic **110**. In one embodiment, optic **110** is a clear optic component configured to direct light from light source **108** (at a proximal end of optic **110**) to blade **112** (at a distal end of optic **110**). In some embodiments, optic **110** can comprise a lens **114**, such as a collimator total internal reflection (TIR) lens, at the proximal end to direct light in a more linear manner distally. This is depicted in FIG. 4, with the horizontal lines in optic **110** representing light from light source **108** being directed by lens **114** from the larger proximal end of optic **110** to the smaller distal end of optic **110**, and then to blade **112**.

(42) Optic **110** can comprise a clear molded material, such as resin, plastic, polycarbonate, acrylic, glass or another suitable material. In one embodiment, optic **110** and lens **114** comprise a unitary molded component, while in other embodiments optic **110** and lens **114** can comprise separate, distinct components mounted adjacent to and optically coupled with one another. As depicted in

FIG. 4, optic **110** has a generally circular cross-section, with a portion that widens from lens **114** and then tapers towards blade **114**, where the cross-section becomes rectangular. In various embodiments, optic **110** can be sized to fit a corresponding LED or other lighting component of light source **108** and have side walls congruent with a TIR surface or lens while also being configured to mate with blade **112**.

(43) Referring also to FIG. 5, the distal end of optic **110** is optically coupled to a proximal end of blade **112**. Blade **112** also can be referred to as an electrode and, in use, can be used to cut or coagulate soft tissue of a patient. In various embodiments, blade **112** can telescope and be locked at a desired length, have a bendable shaft, include suction, and comprise many different tip configurations and designs to accommodate a variety of medical procedures. The optical and illumination features discussed herein can be compatible with some or all of these variations of blade **112**, providing customizability and flexibility desired by surgical users.

(44) Blade **112** comprises a metal or metal alloy, such as steel, in embodiments and is coated with a dielectric material (or insulator), which focuses RF energy applied to blade **112** to achieve incisions with minimal thermal damage to the tissue. In one example, the dielectric material comprises glass. Other example dielectric materials include porcelain or ceramic, mica, plastics, and the oxides of some metals. In embodiments in which blade **112** is coated with glass, the glass can be clear and used as a conduit for light into the surgical site. Thus, the optical coupling of optic **110** with blade **112** enables light from light source **108** to be guided and directed to a distal end of blade **112** via optic **110** and the dielectric coating on blade **112**. From the distal end of blade **112**, the light is directed out of the dielectric coating to illuminate the area around the distal tip of blade **112**, such as a surgical site. This is depicted in FIG. 6.

(45) Referring to FIG. 7, the dielectric coating and blade **112** can comprise at least one light-guiding feature, which can be customized to influence or direct the light in particular directions. In this embodiment, a set of grooves **116** is formed in or on blade **112** and causes light to be directed out of the dielectric coating applied on blade **112** in a particular way, direction or pattern. The location, size, number, and pattern of grooves **116** can vary in other embodiments in order to provide particular light effects. For example, grooves **116** can be linear, arcuate, zig-zag, random or comprise one or more geometric shapes, such as a circle, square, triangle, polygon or other shape. In still other embodiments, features other than grooves, such as raised portions or patterns, additional coatings or layers, or embedded elements, can be incorporated in order to provide desired light effects. Grooves **116** also can comprise combinations of these features or shapes and can be located on a proximal portion **112a** or a distal portion **112b** (or both) of blade **112**. The thickness of the dielectric coating, with or without grooves **116** or other features discussed herein below, can range from about 1 micron to about 100 microns, such as from about 1 micron to about 10 microns, from about 1 micron to about 50 microns, from about 10 microns to about 100 microns, from about 10 microns to about 90 microns, from about 50 microns to about 100 microns, from about 20 microns to about 80 microns, from about 30 microns to about 70 microns, from about 40 microns to about 60 microns, from about 60 microns to about 90 microns, or some other range within the range bounded by 1 micron and 100 microns. In still other embodiments, the thickness of the dielectric coating can be greater than 100 microns.

(46) Referring also to FIG. 8 (which is not necessarily drawn to scale), several examples of patterns or features other than grooves **116** are depicted. These patterns or features, like those discussed with respect to FIG. 7, can be three-dimensional. At (a) in FIG. 8, a prism pattern is shown. While the prism pattern depicted at (a) is regular, in other embodiments it can be irregular, with prisms of varying sizes or heights or interspersed with other shapes or features, such as flat areas or grooves. Systemized patterns other than prisms can be used in other embodiments. At (b), a smooth, highly polished optical quality surface example is shown. In one embodiment, a portion or the entire surface of (b) can be reflective (i.e., mirrored). At (c), a random pattern, such as might result from a surface being grit-blasted or bead-blasted, is shown. One or more of these patterns or features or

those discussed with respect to FIG. 7 can be incorporated in the dielectric layer or on one or more areas of blade **112**, on the entirety of the dielectric layer or blade **112**, or combined with other features or elements on some or all of blade **112** in various embodiments in order to provide a desired light effect.

(47) In still other embodiments, and referring to FIG. 9, blade **112** or the dielectric layer can comprise or have a multi-layered configuration to tailor the optical properties to achieve a desired outcome. In one example, the dielectric layer on blade **112** can be stratified, comprising a first or base layer **120** and a second or top layer **122** on a substrate **118** of blade **112**. In other embodiments, either substrate **118** or the dielectric layer thereon can comprise more or fewer layers. In still other embodiments, layers of other materials in addition to the dielectric layer also can be incorporated.

(48) In an embodiment with multiple layers such as first layer **120** and second layer **122**, the optical properties of the respective layers can vary from one another. For example, each layer **120**, **122** can have optical properties that are altered by adding pigments or composites, or by varying the degree of crystallinity of the material of the layer **120**, **122**.

(49) Example pigments include P4055 blue, manufactured by Prince Minerals, Inc.; P4020 green, manufactured by Prince Minerals, Inc.; and F6333 black, manufactured by Ferro Corporation. Other pigments in these or other colors, or combinations of these or other colors, also can be used.

(50) Example composites include adding a ceramic material, such as one or more of Al.sub.2O.sub.3, ZrO.sub.2, or TiO.sub.2, to first layer **120** or second layer **122** to absorb light and show a white color.

(51) Modifying crystallinity of first layer **120** or second layer **122** can change the way light is reflected off the surface of blade **112**. Increasing crystallinity will increase the diffused reflection, creating a matte finish. Reducing crystallinity will increase the specular reflection, like a mirror, or glossy. In one embodiment, an insulator crystallizing glass at 1.1 vol % crystallinity has a glossness of 108 gloss units (GU). In another embodiment, at 65 vol % crystallinity the glossness is 7.4 GU.

(52) In a particular example related to FIG. 9, first layer **120** can be applied directly to substrate **118** of blade **112** and have absorptive properties, while second layer **122** can be applied onto first layer **120** and have highly transmittive properties. Other combinations of layers with these or other optical properties can be selected and used in other embodiments in order to achieve a desired light-effective result. Each layer need not be applied over an entirety of substrate **118** of blade **112**, such that different portions of blade **112** can comprise more or fewer layers with the same or different optical properties. Additionally, one or more layers of a multi-layered dielectric also can comprise one or more of the features discussed with respect to FIGS. 7 and 8 or other features discussed herein.

(53) Referring to FIG. 10, which depicts a cross-section of blade **112**, in some embodiments blade **112** can comprise additional light-guiding or light-influencing elements in order to achieve desired light-guiding effects. In the embodiment depicted in FIG. 10, and similar to the embodiment of FIG. 9, substrate **118** of blade **112** comprises a multi-layered coating. A base coating **124** is applied onto substrate **118** to provide adhesion and insulation and fulfill mechanical, thermal and electrical requirements of blade **112**, considering that a primary purpose of blade **112**, as part of device **100**, is electrosurgery. A top coating **126** is applied over base coating **124** and, in one embodiment, comprises an optical fiber **128** or other optical element embedded therein. Top coating **126** can be optimized for optical properties (while still meeting dielectric strength and other requirements of electrosurgical device **100**) and can comprise glass or another suitable material, including materials discussed elsewhere herein. Base coating **124** can comprise the same or a different material than top coating **126**.

(54) The process of applying top coating **126** over base coating **124** on blade **112** can comprise firing to fuse optical fiber **128**, which also can comprise glass or pure silica fiber, in top coating **126** or between top coating **126** and base coating **124**. Optical fiber **128** and top coating **126** have

softening temperature requirements such that the temperature should be close to or higher than the firing temperature of base coating **124** so that the embedded optical fiber **128** is not deformed, and is able to trap light within optical fiber **128**. In one embodiment, the firing temperature of base coating **124** is about 800 C, and the softening temperature of optical fiber **128** and top coating **126** is close to or higher than 800 C.

(55) The difference in refractive indices between top coating **126** and optical fiber **128** will result in light being confined to optical fiber **128**. In other embodiments, optical fiber **128** can comprise a glass core and glass cladding and therefore itself have a difference in refractive indices between these layers.

(56) The material(s) of top coating **126** can be selected to meet desired optical requirements or effects. For example, top coating **126** can be colored, such as green or blue, or transparent. In one particular embodiment, top coating **126** can comprise a light absorptive material such that light is emitted only from optical fiber **128** at a distal tip of blade **112** where optical fiber **128** is exposed to air (see FIG. **11** and related discussion). In one embodiment, a light absorptive material comprises black pigment (e.g., F6333 manufactured by Ferro Corporation). An advantage of such an embodiment is that a non-transparent coating will not show the underlying light path through optical fiber **128** along blade **112**, which may be preferred by users. As discussed with respect to FIGS. **7** and **8**, top coating **126** also can comprise grooves **116** or other light-influencing features.

(57) As depicted in FIG. **10**, optical fiber **128** is embedded on only one side of blade **112**. In other embodiments optical fiber **128** can be embedded on both sides of blade **112** or have some other configuration (e.g., embedded only along a central portion of blade **112**, or embedded in an array on one or both sides of blade **112**). For example, and referring to FIG. **11**, a plurality of optical fibers **128** can be embedded in coating **126** in an arrangement that results in light emitted at the distal tip of blade **112**. In such an embodiment, base coating **124** can comprise glass, and top coating **126** can comprise a colored dielectric glass. A similar embodiment with an alternate arrangement of optical fibers **128** is depicted in FIG. **12**. In FIG. **12**, optical fibers **128** are arranged in a grid-like pattern such that light is emitted around an entire cutting edge of the distal tip of blade **112**. Such a configuration can give a surgeon user a more precise view of a cutting location regardless of the orientation of blade **112** during use. Various other arrangements of optical fibers **128** can be implemented in other embodiments in order to provide a desired lighting effect, as appreciated by those of skill in the art.

(58) Another embodiment of an illuminated electrosurgical device **200** is depicted in FIG. **13**. Device **200** comprises a handpiece **202** (like numbers are used herein throughout, incremented by 100, unless otherwise discussed), blade assembly **204**, switch assembly **206**, cable assembly **207**, suction tubing assembly **209**, and finger grip **211**. Handpiece **202** comprises a housing **203**, which houses blade assembly **204**, switch assembly **206**, and other components of device **200**. Device **200** can comprise additional components, some of which are depicted but not particularly discussed.

(59) With respect to illumination in particular, and referring also to FIG. **14**, blade assembly **204** comprises a ring lens **230** arranged around a central shaft **232** of blade assembly **204**, coupled to a hub **234**. Ring lens **230** is arranged at a distal end of handpiece **202**, centered on central shaft **232**. Blade assembly **204** also comprises a lock nose cone **236** that couples to hub **234**.

(60) Ring lens **230** is depicted in greater detail in FIGS. **15-19**. In the depicted embodiment, ring lens **230** comprises a Fresnel lens. A typical Fresnel lens comprises a series of facets or prisms that focus or magnify light. These facets can be seen in FIGS. **15** and **16**. FIG. **15** depicts one configuration of facets on a front side of ring lens **230**, while FIG. **16** depicts one configuration of facets on a back side of ring lens **230**. Ring lens **230** can comprise acrylic, polycarbonate, or some other transparent material. Surfaces of ring lens **230**, such as the major surfaces visible in FIGS. **15** and **16**, can be textured, such as according to texture grade MT-11006 in one example. In some embodiments, ring lens **230** can comprise total internal reflection (TIR) surfaces, which can maximize the light that is first reflected within and then exited from ring lens **230**. Ring lens **230**

also can be considered to be a light pipe.

(61) In device **200**, ring lens **230** can focus light from a light source (discussed herein below) and direct the light around a tip of blade **212** of blade assembly **204**. The dimensions of ring lens **230** and configuration of the facets thereof can be optimized to provide a particular illumination effect. In one embodiment, ring lens **230** has a diameter of approximately 0.78 inches and is about 0.2 inches thick. Such a ring lens **230** can provide an approximately 2-inch circle of light around a distal tip of blade **212** in use. The diameter can vary from about 0.5 inches to about 1.0 inches, and the thickness can range from about 0.1 inches to about 0.25 inches. Adjusting the size or dimensions of features of ring lens **230** can provide other illumination effects, as may be desired in particular applications or uses. Dimensions can be selected such that an outer diameter of ring lens **230** does not block a user's view while at the same time being large enough to provide a light angle that minimizes shadow at the tip of blade **112**.

(62) Light can be provided by one or more light emitting diodes (LEDs) arranged on an LED printed circuit board (PCB) **240** of cable assembly **207**, which is depicted in FIGS. **13** and **20**. The LEDs can comprise chip on board (COB) LEDs, which can comprise a plurality of LED chips. COB LEDs can take up less space and be bonded directly to a substrate, which can be LED PCB **240**. LED PCB **240** can comprise a partial or full ring configuration, which can be sized and arranged such that COB LEDs mounted thereon provide light directly to ring lens **230**.

(63) LED PCB **240** is coupled to a light handle PCB **242**, which includes contacts **244** and **246** for switch assembly **206**. In particular, contacts **244** electrically couple with a main button pad **248** of switch assembly **206**, and contact **246** electrically couples with a light switch **250** of switch assembly **206**. In one embodiment, main button pad **248** comprises one or more PANASONIC® EVQQ2 switches by which device **200** can be turned on and off and otherwise controlled, and light switch **250** comprises a slider switch by which illumination of device **200** can be turned on and off. Other types of switches can be used in other embodiments. In another embodiment, illumination can be automatically turned on or off when device **200** is turned on or off, respectively. In still another embodiment, device **200** must first be turned on via main button pad **248** before illumination can be turned on via light switch **250**. Wires **252** electrically couple LEDs on LED PCB **240** to light handle PCB **242** and light switch **250**.

(64) Referring also to FIG. **21**, LEDs on LED PCB **240** or device **200** overall can be powered by an external power source or an internal battery **258**. An internal battery can be disposable and replaceable, or rechargeable. In one embodiment, at least one lithium CR-123 battery is housed in cable assembly **207** and coupled to LED PCB **240** via light handle PCB **242**, which is in turn coupled to battery control circuitry on a battery control PCB **254** by a cable **256**. Battery **258** can be accessed via a battery door **260** in a connector **262** of cable assembly **207**. Batteries other than lithium or CR-123 can be used in other embodiments.

(65) As previously mentioned, LED PCB **240** can comprise a partial or full ring configuration that is sized and arranged such that COB LEDs mounted thereon provide light directly to ring lens **230**. This can be seen in FIGS. **22-26**. The embodiment of FIGS. **22-24** comprises two LEDs **270**, while the embodiment of FIG. **25** comprises four LEDs **270**. FIG. **26** is an enclosed housing view of device **200** and can comprise two, four or some other number of LEDs **270**.

(66) Referring in particular to FIG. **24**, in operation light from LEDs **270** is directed into ring lens **230**, which concentrates the light into a circle around a distal tip of blade **212**. In the embodiment of FIG. **24**, light is concentrated into a circle with a diameter of about 2 inches, though in other embodiments this diameter can be made larger or smaller, or comprise a different shape, by adjusting properties of ring lens **230**.

(67) The same principles apply to the embodiment of FIG. **25**, which comprises four LEDs **270**. As can be seen in FIG. **25**, each of the four LEDs **270** can be evenly spaced apart (e.g., by about 90 degrees) from one another along a circumference on LED PCB **240**. Other embodiments can comprise more or fewer LEDs **270**, which can be similarly arranged with even spacings along a

circumference on LED PCB **240** or in some other arrangement chosen, for example, to provide a particular illumination effect.

(68) Lens **230** also can be differently configured in other embodiments to provide a particular light effect in cooperation with one or more LEDs **270**. For example, in FIG. **27** lens **230** can comprise two legs **272a** and **272b**, each leg **272a**, **272b** configured such that a proximal end is arranged relative to a respective LED **270** to receive light therefrom. The proximal end of each leg **272a**, **272b** can comprise a collimator in the form of two TIR surfaces to reflect light internally to lens **230**, thereby concentrating and controlling the light to provide illumination in a desired pattern around the distal tip of blade **212**. In embodiments in which device **200** comprises more or fewer LEDs, lens **230** can correspondingly comprise more or fewer legs **272a**, **272b**. Still other configurations of lens **230** are possible, as appreciated by those of skill in the art. For example, different shapes of lens **230** or legs **272a**, **272b** can be implemented, as can different configurations of TIR surfaces on legs **272a**, **272b** or lens **230**.

(69) In yet another embodiment, device **200** can comprise additional components or features to provide increased or desired illumination effects. For example, device **200** can comprise additional collimating features or lenses. In another example, LEDs **270** can comprise or be configured for particular light effects, such as comprising a collimator or other feature on the significant surface thereof to direct light. In yet another example, LEDs **270** can be arranged on LED PCB **240** or otherwise positioned in light-effective ways, such as at an angle, including orthogonally. In still another embodiment, a color or colors of LEDs **270** can be selected to provide brighter or more desired light, such as white light. LEDs **270** also may be arranged to be user-manipulated in some embodiments, such as to be brightened or dimmed (intensity variation) or mechanically moveable, such as closer or further from the distal tip of blade **212** or relative to lens **230** in order to tune a size or other characteristic of the light directed towards the distal tip of blade **212**.

(70) Several advantages are provided by embodiments discussed herein. These include compatibility of ring lens **230** of the illumination system with other features of electrosurgical device, such as extendability and rotation, as well as a bendable tip, of blade assembly **204**, and smoke evacuation. This is possible because of the ring configuration of ring lens **230**, which enables other components to pass through.

(71) Features and components of different embodiments discussed herein can be combined in other embodiments. For example, blade coatings and optical fibers discussed with respect to device **100** can be incorporated into embodiments of device **200**, and vice-versa. In this was particular illumination effects can be designed and achieved in order to meet particular desires or needs in the industry.

(72) Various embodiments of systems, devices, and methods have been described herein. These embodiments are given only by way of example and are not intended to limit the scope of the claimed inventions. It should be appreciated, moreover, that the various features of the embodiments that have been described may be combined in various ways to produce numerous additional embodiments. Moreover, while various materials, dimensions, shapes, configurations and locations, etc. have been described for use with disclosed embodiments, others besides those disclosed may be utilized without exceeding the scope of the claimed inventions.

(73) Persons of ordinary skill in the relevant arts will recognize that the subject matter hereof may comprise fewer features than illustrated in any individual embodiment described above. The embodiments described herein are not meant to be an exhaustive presentation of the ways in which the various features of the subject matter hereof may be combined. Accordingly, the embodiments are not mutually exclusive combinations of features; rather, the various embodiments can comprise a combination of different individual features selected from different individual embodiments, as understood by persons of ordinary skill in the art. Moreover, elements described with respect to one embodiment can be implemented in other embodiments even when not described in such embodiments unless otherwise noted.

(74) Although a dependent claim may refer in the claims to a specific combination with one or more other claims, other embodiments can also include a combination of the dependent claim with the subject matter of each other dependent claim or a combination of one or more features with other dependent or independent claims. Such combinations are proposed herein unless it is stated that a specific combination is not intended.

(75) Any incorporation by reference of documents above is limited such that no subject matter is incorporated that is contrary to the explicit disclosure herein. Any incorporation by reference of documents above is further limited such that no claims included in the documents are incorporated by reference herein. Any incorporation by reference of documents above is yet further limited such that any definitions provided in the documents are not incorporated by reference herein unless expressly included herein.

(76) For purposes of interpreting the claims, it is expressly intended that the provisions of 35 U.S.C. § 112(f) are not to be invoked unless the specific terms “means for” or “step for” are recited in a claim.

Claims

1. An electrosurgical device comprising: an electrosurgical blade having a proximal end and a distal end and comprising a dielectric coating, at least one of the electrosurgical blade or the dielectric coating including at least one light-guiding element comprising at least one groove formed on the electrosurgical blade or the dielectric coating at a location proximal of the distal end of the electrosurgical blade, the at least one groove oriented to direct light through the dielectric coating towards the distal end of the electrosurgical blade, wherein the dielectric coating includes the at least one light-guiding element; an optic having a proximal end and a distal end, the distal end of the optic optically coupled to the proximal end of the electrosurgical blade; and a light source arranged at the proximal end of the optic and configured to provide light through the optic and into the dielectric coating, the dielectric coating and the at least one light-guiding element configured to guide light from the optic through the dielectric coating, and out of the dielectric coating near the distal end of the electrosurgical blade to illuminate an area around the distal end of the electrosurgical blade.
2. The electrosurgical device of claim 1, wherein the light sources comprises at least one light-emitting diode (LED).
3. The electrosurgical device of claim 1, wherein the optic comprises a lens at the proximal end of the optic.
4. The electrosurgical device of claim 3, wherein the lens is a collimator total internal reflection (TIR) lens.
5. The electrosurgical device of claim 1, wherein the dielectric coating comprises glass.
6. The electrosurgical device of claim 1, wherein the illuminated area around the distal end of the electrosurgical blade has a diameter of about two inches.
7. The electrosurgical device of claim 1, wherein the at least one groove is formed on the electrosurgical blade.
8. The electrosurgical device of claim 1, wherein the at least one groove is formed on the dielectric coating.
9. An electrosurgical device comprising, an electrosurgical blade having a proximal end and a distal end and comprising a dielectric coating, at least one of the electrosurgical blade or the dielectric coating including at least one light-guiding element comprising at least one pattern of texture formed on the electrosurgical blade or the dielectric coating at a location proximal of the distal end of the electrosurgical blade, the at least one pattern of texture oriented to direct light through the dielectric coating towards the distal end of the electrosurgical blade, wherein the dielectric coating includes the at least one light-guiding element; an optic having a proximal end and a distal end, the

distal end of the optic optically coupled to the proximal end of the electrosurgical blade; and a light source arranged at the proximal end of the optic and configured to provide light through the optic and into the dielectric coating, the dielectric coating and the at least one light-guiding element configured to guide light from the optic through the dielectric coating, and out of the dielectric coating near the distal end of the electrosurgical blade to illuminate an area around the distal end of the electrosurgical blade.

10. The electrosurgical device of claim 9, wherein the dielectric coating is a multi-layered dielectric coating comprising a base layer and a top layer.

11. The electrosurgical device of claim 10, wherein at least one of a pigment, a composite content or a degree of crystallinity varies from the base layer to the top layer.

12. The electrosurgical device of claim 10, wherein at least one optical element is embedded in the multi-layered dielectric coating.

13. The electrosurgical device of claim 12, wherein the at least one optical element is an optical fiber.

14. The electrosurgical device of claim 12, wherein an array of optical fibers are embedded in the multi-layered dielectric coating.

15. The electrosurgical device of claim 9, wherein the pattern of texture comprises raised features.

16. The electrosurgical device of claim 9, wherein the pattern of texture comprises prisms.

17. The electrosurgical device of claim 9, wherein the pattern of texture is random.

18. The electrosurgical device of claim 9, wherein the pattern of texture is formed on the electrosurgical blade.

19. The electrosurgical device of claim 9, wherein the pattern of texture is formed on the dielectric coating.

20. A method comprising: coating an electrosurgical blade having a proximal end and a distal end with a dielectric layer; forming at least one light-guiding element on at least one of the electrosurgical blade or the dielectric layer at a location proximal of the distal end of the electrosurgical blade including forming at least one groove on the electrosurgical blade or the dielectric coating as the at least one light-guiding element, the at least one groove oriented to direct light through the dielectric coating towards the distal end of the electrosurgical blade; optically coupling a distal end of an optic to the proximal end of the electrosurgical blade; and arranging a light source at a proximal end of the optic to provide light through the optic and into the dielectric coating, the dielectric coating and the at least one light-guiding element configured to guide light from the optic through the dielectric coating, and out of the dielectric coating near the distal end of the electrosurgical blade to illuminate an area around the distal end of the electrosurgical blade.

21. The method of claim 20, further comprising forming the optic to include a lens at the proximal end of the optic.

22. The method of claim 21, wherein arranging a light source further comprises arranging the light source relative to the lens of the optic such that total internal reflection (TIR) of the light occurs in the optic.

23. The method of claim 20, wherein coating an electrosurgical blade with a dielectric layer further comprises forming at least one optical fiber in the multi-layered dielectric as the at least one light-guiding element.

24. The method of claim 20, wherein forming at least one light-guiding element further comprises forming at least one pattern on the electrosurgical blade.

25. The method of claim 20, wherein forming at least one light-guiding element further comprises forming at least one pattern on the dielectric coating.
